

Synthesis: Causal Models, Causal Knowledge

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This article presents the 2018 Linda Crane Memorial Lecture Award on Synthesis: Causal Models, Causal Knowledge. The synthesis is that practice is based on causal knowledge and that we can encode causal knowledge with causal models. It proposes the use of causal models to provide a synthesis of what we know to develop causal knowledge from which to further develop with empirical evidence and from which knowledge we can practice. The synthesis is founded on critical realist epistemology that explicitly identifies the human mind as part of the process of knowledge and where the models that are constructed fit with reality even when we cannot empirically verify every component of such models; but where knowledge assumptions are encoded in the models clearly, combine knowledge with reasoning and are subject to empirical verification when possible. (*Cardiopulm Phys Ther J.* 2018;29:134–143) **Key Words:** *epistemology, causal models, knowledge based practice, differential diagnosis, critical realism*

This article presents the 2018 Linda Crane Memorial Lecture. Readers will find it weaves a personal story with a theoretical thesis and an attempt to provide a synthesis. The synthesis is that practice is based on causal knowledge and that we can encode causal knowledge with causal models. These ideas have been in development for nearly my entire career, but my pursuit picked up pace in 2005 following the publication of my dissertation.

The Linda Crane Memorial Lecture is named and exists because of Dr. Linda Crane's legacy and contributions to physical therapy practice and education. Please consider reading through the Linda Crane Lectures, published each year and available online through the Cardiopulmonary PT Journal Web site since 2000 when Cohen¹ presented the first Linda Crane Lecture in February, at CSM in New Orleans. Individually and in sum, they offer an amazing testament and historical sweep of Dr. Crane's life and legacy, as well as the changing times of physical therapy these past 19 years.

Despite never having had the opportunity to meet Dr. Crane, the fruits of her labor did not escape my early career. In the late 1990s, her work on competencies were critical to me taking over the responsibility for the cardiopulmonary physical therapy course at the UMass Lowell and studying for the specialization examination. The very idea of competencies was new to me. Dr. Crane's focus, decisiveness, and passion was an inspiration.

Reading through the corpus of the Linda Crane lectures reminded me of a paragraph from Taylor's² book, *The Myth of Certainty: The Reflective Christian and the Risk of Commitment*. This paragraph resonates with me and resonated with my reading about Dr. Crane as a professional who was humble, decisive, and passionate:

"Humility is not synonymous with passivity or indecisiveness. One can hold beliefs passionately yet with humility. An awareness of the slipperiness of truth, the subtlety of error, and the human appetite for illusion does not mean one cannot believe and act with intensity. Acting with less than perfect knowledge is part of the risk of being human. But we should therefore reject self-congratulating narrowness, always seeking a deepening and broadening of our understanding rather than a hardening of it." (p. 127, emphasis added)

This article is about the fact that acting with less than perfect knowledge is part of the risk of being human. The goal is to get you to think about knowledge-based practice

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(KBP). A way of thinking that tries to balance the fact that acting with less than perfect knowledge is part of the risk of being human with the fact that in the clinical setting, everything we do is an action (even inaction is an action) based on a decision based on less than perfect knowledge. Let's call this the clinician's dilemma.

EPISTEMOLOGICAL STARTING POINT: COMMON SENSE REALISM

Before we go too far, we should agree with a set of axioms about knowledge, which means we are doing epistemology. Epistemology is the branch of philosophy concerned with knowledge, especially regarding its methods, validity, and scope. It is the investigation of what distinguishes justified true belief (as a definition of knowledge) from opinion. My epistemology is influenced by my Christian worldview and tends toward McGrath's explication of Bhaskar's ideas, an epistemology called critical realism.^{3,4} If you have read any of my early editorials, you are familiar with some of the implications of being a critical realist.^{5,6} In one such editorial, readers were reminded to "keep in mind that what you can prove is easy to know, but what you know may be exceptionally difficult to prove."⁵ While not exclusive to critical realism, that statement is an implication of the critical realist approach to epistemology (more implications will come later).

You may be an empiricist, a rationalist, a pragmatist, maybe an idealist, some of you may not think about your epistemology or worldview. But, whether you think about it or not theoretically, you are in fact practicing some form of epistemology practically, which may or may not cohere with your worldview. It is important to have a basic set of axioms when communicating knowledge. A set of axioms that we can all agree on regarding the methods, validity, and scope of knowledge. I have found it helpful to share the basic axioms of common sense realism in each of my courses and ask students to let me know if there are any points of disagreement. This allows us to start off with at least a basic foundational set of agreements about how the world works and what we can know of it. Common sense realism is foundational to many epistemological systems and therefore should not be averse to empiricists, scientific realists, or even pragmatists. My education on these is from the philosopher, theologian, apologist, and preacher Sproul,⁷ who learned them from the writings of philosopher and Reverend Reid.⁸ If we do not agree with these, then we are likely going to have a hard time in the teaching—learning process since the teaching—learning process is about sharing knowledge.

The *Law of Non-Contradiction* states that something cannot be and not be in the same way at the same time. Violation of noncontradiction raises the flag that something is wrong. The *Law of Causation* states that every event that is had a cause. It doesn't mean we know what it is, just that a cause resulted in the observed effect. The *Basic reliability of sense perception* says that we agree that what we sense and perceive is basically reliable. There are

occasional illusions and discrepancies between observers, but they are basically reliable and with our limitations we are capable of understanding reality. The *Analogical use of language* says that to talk about knowledge, we must use language. The words we use are not actually the things they represent. There is an ontological distinction between the muscle and the word muscle (ontology refers to the branch of philosophy that deals with the nature of being). But we use the word muscle to represent muscle. We must assume this works from the start—we must assume we can use language as an analogy of something real that is ontologically distinct from the word itself.

An example that weaves some of these basic axioms together would be teaching a lab on lung sounds. We teach about crackles (a certain sound we hear that is not normally present with auscultation of the lungs). There seem to be at least 2 distinct types of crackle sounds. Therefore, we have a word to describe the analogical representation of sounds—fine or coarse crackles. These words are analogically related to the ontological entity (a sensory perception in our brain). Whether we hear fine crackles or coarse crackles is connected to our sensory perception and then we utter the words to represent what we hear.

But sometimes people use the words dry crackles when they hear fine crackles. They are using a different word to modify their crackles. The words dry crackles represent something different ontologically. The words dry crackles are going beyond the sensory perception of sound to something about the nature of the cause of the sound. A dry crackle is presumed to be caused by a dry source—something that is not wet, not mucous, not secretions, not edema, not inflammation. When discussing what we know, the use of the language dry crackle reports that we know the cause of the sound we hear (or at least some general characteristic of the cause). Whereas a fine crackle merely reports on the sound we have perceived and, as such, is analogically related to only the effect (not the cause) and therefore contains less knowledge. A question would be whether such a claim is justified? That is related to the ontology of the cause—effect relationship of dry crackles. If the cause of fine crackles is only ever a dry cause, then we probably are justified since fine crackles could only be the effect of a dry cause. I am not making that claim here—just raising it as a point.

Similarly, there is more knowledge in the statement: "That patient has angina." Than the statement: "That patient has chest pain." Even though we recognize that angina is often chest pain, chest pain is not always angina. Why? These words are referring to different real things ontologically. Chest pain is referring to pain in the chest (an effect) with no claim about the cause of that pain. But angina is referring to a symptom (usually pain in the chest) that, for some reason, we feel justified in claiming to know the cause of that pain (myocardial ischemia). It is typical (or should be) to withhold the term angina for times when additional information brings additional knowledge that makes abduction to ischemia the best explanation of the chest pain prior to using the term angina.

KNOWLEDGE-BASED PRACTICE

One of the Dr. Crane's contributions to our profession was competency. Let's connect KBP to competency. Competence requires knowledge. Therefore, practice competence requires practice knowledge. These statements are equivalent to the implication: If practice knowledge, then practice competence. Whether this implication represents a true causal relationship could be up for discussion. For instance, we may simply be using different words analogically to represent the same thing ontologically. But for the sake of this article, let's consider practice competence and practice knowledge as ontologically distinct; let's further claim that practice knowledge is a necessary but not a sufficient condition for practice competence. Practice knowledge is not sufficient because we believe that other factors would have an influence on practice competence and therefore could interfere with the relationship between practice knowledge and practice competence. To be clear and up until now, we have the very uninteresting and uncontroversial statement that practice knowledge is a necessary but not a sufficient condition for practice competence. This article claims that KBP can help set the stage for practice competence.

This article unpacks the potential value of these approaches. Thus far my writing on this topic has taken the form blogs,⁹ editorials,^{5,6} presentations,¹⁰ and the development of a new DPT program that equips students with the tools—as developing clinicians—to practice physical therapy and to develop their practice through experiences and the interpretation of research studies using causal models.¹¹

Knowledge-based practice includes synthesis, building causal models, generating inference rules for these models for practice, and testing the models—and doing so explicitly and iteratively as part practice, research, and education. A core concept of KBP is that the development of causal models, once the language and methods are established and disseminated, provides a bridge for knowledge development in the profession; as opposed to evidence that identifies empirical connections that is then translated to the clinician.

Recall the clinician's dilemma that acting with less than perfect knowledge is part of the risk of being human. Evidence-based practice (EBP) won't work. It will not resolve the clinician's dilemma. It will not create perfect knowledge. But let me put this in context. First, I am characterizing EBP. No one claims that EBP will solve the clinician's dilemma, and no one claims that EBP will lead to perfect knowledge. And, it is quite possible that you will read this article about KBP, cause, models, and inference and say that it simply is EBP repackaged or that it is knowledge translation with some tools borrowed from epidemiology.

These are all possible claims and points of further discussion. The ideas being shared have been in development since about 2005. This was a big year for me—having defended my dissertation in April 2003, and

having taken the highly recommended “go easy and decompress” year following my dissertation included the question of what to do next in my academic career.

But going back even further, there were early experiences that put me on this path, including my early experience as a PT student in 1992. We were told we had to pursue the MS in Physical Therapy to improve the profession's ability to generate and use evidence for practice. But we were also told that answers to questions could not be answered and that we would eventually know the answer as we gained practice experience. These 2 statements are in tension. And it took me many years to realize the tension, its cause, what it meant, and what we might be able to do about it.

My doctoral training in ergonomics and epidemiology equipped me with an understanding of empirical methods. How to examine cause–effect relationships within complex dynamic systems; how to characterize variables for exposure modeling and how to conduct, interpret, and apply mathematical and statistical analysis to observations of reality, and with an understanding of theory building. In other words, scientific realism including the implicit interplay between empirical and rational epistemologies.

After my dissertation was completed and being at a loss for what it all meant, my wife took me to church. Growing in faith opened me to the intellectual world of theology, philosophy, and apologetics with formal and self-study. With this came a familiarity with a broader perspective of the ways of knowing—epistemology; the formal rules of inference via deduction, induction, and abduction through logic, and common sense realism as a foundational epistemology.

For example, in “How to Read a Book” by Adler and Van Doren,¹² his final level of reading called: syntopical reading, which synthesizes. It is complex and systematic; the reader compares one book with another and with other sources. The syntopical reader can construct new understanding that does not appear in any of the other books. The syntopical reader is constantly thinking and questioning while reading, making connections to gain understanding, the reader develops knowledge through synthesis.

It was with this background and in this context that I read Alan Jette's 2003 invited commentary¹³ where he posed the question: “What can be done to stimulate more research in physical therapy that has direct clinical relevance?” My editorial response proposed that this was a challenging task because “the simple study of parts in isolation—reductionism—is the *modus operandi* of the scientific method, attempting to isolate sources of variation. Clinicians, however, are faced with all sources of variation at the same time and must deal constantly with the full burden of the complex system.”¹⁴ From then onward, my work on this question has continued, particularly how to accomplish what was being proposed in that editorial.

EPISTEMOLOGICAL FOUNDATION: CRITICAL REALISM

McGrath, in his *Scientific Theology, Volume 2: Reality*, introduced me to a valuable dialogue partner in Bhaskar in his 1975 book *A Realist Theory of Science* where he expounds his transcendental realism, which has become more commonly known now as “critical realism.”^{4,15} Critical realism is the epistemological foundation for a KBP, which is contrary to the more empirically founded EBP.¹⁶ Critical realism takes common sense realism and scientific realism and points out important considerations about reality and knowledge and opens up opportunities for the discovery of causal associations and their connection into causal models.

McGrath provides a nice concise definition of critical realism: “Reality is apprehended by the human mind, which attempts to express and accommodate that reality as best it can with the tools at its disposal—such as mathematical formula or mental models.”³ Importantly, as an epistemological foundation for a KBP: critical realism explicitly identifies the human mind as part of the process of knowledge. This makes explicit use of models to represent the reality encountered. In other words, the mental models that are constructed are knowledge, and their goal is to fit with reality even when we cannot empirically verify every component of such models. Reality orders what we encounter. Therefore, reality frames our mental models, and empirical evidence frames and constrains our mental models. But they are mental models. And therefore, will lead to inferences that extend beyond what can be observed while still being representative of the real, after all our powers of observation are limited (note the basic reliability in sensory perception).

An important aspect of critical realism is that it claims that ontology determines epistemology: which the way things are determines the way things are known. Ontologically, there is a differentiation and stratification of reality that must be recognized. We need to keep in mind from the idea that ontology determines epistemology that, quoting McGrath: “The nature of reality is such that certain things can only be known to a certain extent, and in a certain way—and that this is the reality of the situation. We are not in a position to determine whether and how things may be known: that is decided by the things themselves.”³

The way practice is should determine the way we know about practice. We practice based on knowledge that is generated from an accumulation of causal models (explicit or implicit), which includes making inferences that extend beyond what can be observed while still being representative of the real, but are generated from universals developed from evidence gathered during structured observations that are guided by particular events; which are events we see while we practice. The way practice is (a use of knowledge in the cause-effect processing of particular events) should determine the way we know about practice (building causal models that are complex

networks of cause–effect relationships to help us understand cause–effect events).

It is not difficult to imagine a clinical researcher claiming, “Your practice isn’t evidence based.” And a clinician claiming, “Your evidence isn’t practical.” This caricature might resonate for a variety of reasons. I don’t mean to imply that all practice is not evidence based, nor that all research is not practical. The point here is to make us think about the extent practice can be evidence based and the extent to which research can be practical.

In other words, how far apart are research and practice? What is the bridge between them? If you believe the bridge is knowledge translation—and I cannot imagine anyone would disagree with a need for knowledge translation—then I simply ask you to consider what the philosophical foundations are for knowledge translation. If knowledge translation is acknowledging that we practice based on knowledge, then we agree. But if knowledge translation is a way to make scientific realism and its occasionally, hyperempirical epistemology practical, then we disagree. But if knowledge translation acknowledges that we practice based on knowledge, then why not acknowledge the transformation from evidence, to universals to causal models to knowledge and simply say we advocate for a KBP and stop using the term EBP?

CAUSAL MODELS AND INFERENCE

When the scientist thinks of evidence, they are often thinking of empirically reduced cause–effect relationships, where by design they “isolate sources of variation.”¹⁴ For example, in a systematic review of whether neuromuscular electrical stimulation (ES) improves maximal oxygen consumption (VO_{2max}), 6-minute walk distance (6MWD), or health-related quality of life (HRQOL) in patients with heart failure (HF), the directional hypotheses are tested as direct causal connections.¹⁷

1. Whether ES causes a change to VO_{2max} ($ES \rightarrow VO_{2max}$)
2. Whether ES causes a change to 6MWD ($ES \rightarrow 6MWD$)
3. Whether ES causes a change to HRQOL ($ES \rightarrow HRQOL$).

However, the clinician is working with many competing hypotheses (Fig. 1), and the clinician is “faced with all sources of variation at the same time and must deal constantly with the full burden of the complex system.”¹⁴ My claim is that the models can be joined; they should be joined; in fact, in epidemiology, that is exactly what is attempted. The program used to make the model in Figure 1 (DAGitty) is freely available, interfaces with the R statistical analysis platform (freely available) and is designed for epidemiologists to plan and interpret studies.¹⁸

Let’s take a closer look at the many causal assumptions of the model in Figure 1. As a causal model, the first thing to realize is that all edges in the graph with an arrow encodes the knowledge that one variable causes the other. This does not have to be a definite causal connection; in fact, most of them are probabilistic and can be stated as

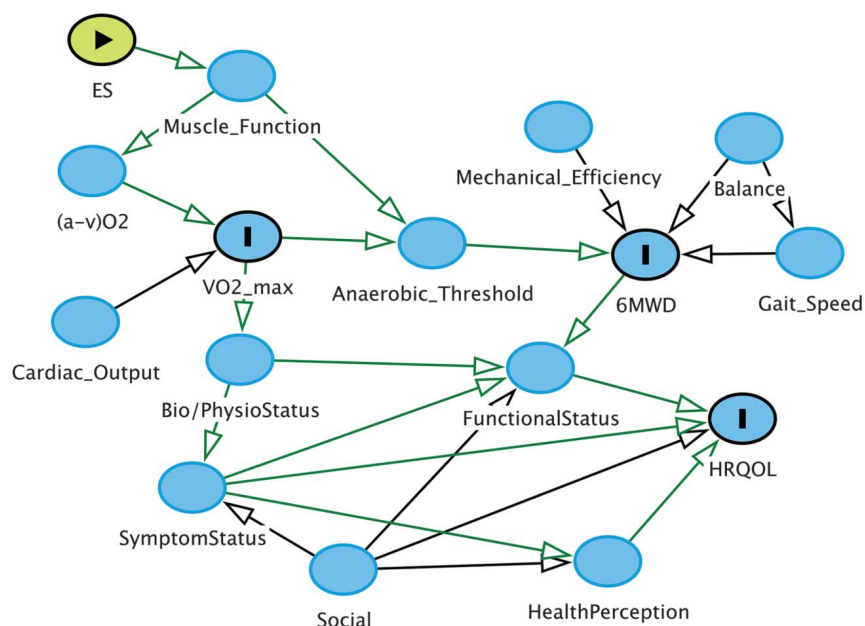


Fig. 1. Possible causal model relating electrical stimulation to possible outcomes. (a-v)O₂, arteriovenous oxygen difference; 6MWD, 6-minute walk distance; HRQOL, health-related quality of life.

conditional probabilities. For example, this graph encodes the knowledge that ES acts as a cause on muscle function. Which can be stated as a conditional probability: the probability of improved muscle function given ES is greater than the probability of improved muscle function given no ES.

The characterization of the intervention (exposure, cause) in this model is discrete (yes/no), but it does not need to be discrete; it can be continuous and can include any of the considered important parameterizations of ES. Also, the effect muscle function is discrete but can be continuous and include any of the important parameterizations of muscle function. In other words, the causal model can encode as much of the ontological information about reality (its variables) as the user would have it encode.

The mathematical and logical implications of the causal model go on to include the multivariable considerations such as the chain rule of conditional probabilities (VO₂max), identification of confounders (balance as a confounder), blocking variables (anaerobic threshold), and adjustment sets (6MWD).

My point here is to answer the question—isn't this the same as concept map? No. Causal models depicted as directed acyclic graphs are based on graph theory and adhere to a set of logical and mathematical rules that allow logical and mathematical implications to be proposed and tested. But they do share concepts. We could say that all causal models are concept maps, but not all concept maps are causal models; therefore, they are not equivalent since equivalence implies bidirectional implication.

This seems like a good point for me to share a rather important fundamental belief that I hold from which this approach hinges that you may have figured out by now. I believe that all practice decisions are—at their core—based

on causal assumptions. I have even come to believe this for ethical practice decisions (this has taken me more years of thinking). Whether you adhere to a deontological (duty based) ethic or a teleological (outcome based) ethic, there are cause-effect assumptions underlying those beliefs and decisions. For example, because I am a professional I must behave this way; or because of this authority, I must not do X. Or because Y will occur, I should not do X. If anyone can think of a practice decision that is not based on a causal assumption, please let me know.

Two more things about the model in Figure 1. First, the lack of an arrow between 2 variables may provide more information than the existence of an arrow. And lastly, this model is developed over time through inductive inferences—the observation of a conjunction of events. It can then be used as we deductively infer possible future states of the world or as we consider the prior state of the world that has led to its current state through abductive inferences.

Figure 2 is expansion of the model in Figure 1 with additional interventions. This model includes aerobic training, resistance training, ES, adaptive equipment, inspiratory muscle training, all as possible interventions for improving 6MWD, VO₂max, and HRQOL in people with HF. No single study has confirmed these causal connections all at the same time. But, a corpus of studies has in fact tested these causal associations. As we learned from McGrath³ about critical realism: “Reality is apprehended by the human mind, which attempts to express and accommodate reality as best it can with the tools at its disposal, such as mathematical formula or mental models.” Critical realism explicitly identifies the human mind as part of the process of knowledge. This makes explicit use of models to represent the reality encountered. In other words, the mental models that are constructed are

knowledge and their goal is to fit with reality even when we cannot empirically verify every component of such models.

Meaning, the model in Figure 2 represents knowledge for practice; practicing based on this model is an example of KBP. I understand that via transitivity, someone would want to say that since evidence leads to knowledge and knowledge informs practice, then evidence informs practice. However, keep in mind that as a critical realist, I do not believe that evidence translates to knowledge directly; I believe that evidence contributes to the transformation of knowledge as something that is ontologically distinct from what evidence can translate to.

Recall from McGrath³: Models that are constructed are knowledge and their goal is to fit with reality even when we cannot empirically verify every component of such models. Let's go back to the situation of the clinician: faced with the dilemma of acting with less than perfect knowledge and faced with all sources of variation at the same time while dealing constantly with the full burden of the complex system. Let's consider the complexity with the simple approach as the number of possible comparisons or alternatives.

If considering 2 interventions with 2 possible states—yes/no (meaning do the intervention or do not do the intervention), there are 2^2 or 4 possible alternatives. If we consider 3 interventions with 2 possible states, there are 2^3 or 8 possible alternatives. This is literally exponential growth of alternatives. If 4 interventions, 2^4 or 16 possible

alternatives; 2^5 is 32; 2^6 is 64; 2^7 is 128. But why encode our 7 interventions as yes/no; why not hi/lo/no with 3^7 or 2187 possible alternatives? How about 4, hi/medium/lo/no for 4^7 or 16,384 possible alternatives? Now that is a dilemma of the complex system and a purely empirical approach will not get us out of that dilemma.

THREE CLAIMS, 2 PREMISES, AND 3 EXAMPLES

I would like to make 3 claims, 2 premises, and provide 3 examples to finish the article.

Three Claims

Claim 1. Causal inferences are the most pervasive in all clinical reasoning.

Implication. When planning your practice and teaching about practice, give clear explicit thought to the causal inferences.

Claim 2. Clinicians reason from cause to most likely effect when prescribing interventions and from effects to most likely cause during their evaluation.

Implication. Consider the incredible limitations in forecasting forward with a limited set of causal factors and the limitations associated with abductive inferences as you reason from observed effect to possible causes.

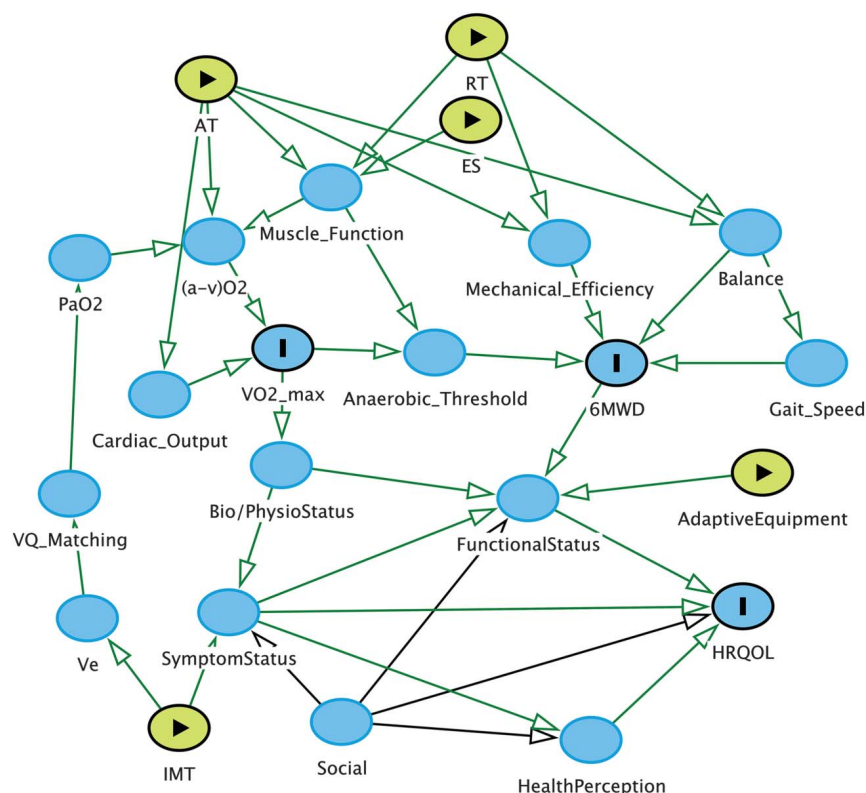


Fig. 2. Expanded intervention causal model relating electrical stimulation to possible outcomes. (a-v)O₂, arteriovenous oxygen difference; 6MWD, 6-minute walk distance; AT, aerobic training; HRQOL, health-related quality of life; IMT, inspiratory muscle training; PaO₂, partial pressure of oxygen in the arterial blood; RT, resistance training; Ve, minute ventilation; VQ matching, ventilation perfusion matching.

Claim 3. Causal inferences are central, either explicitly or implicitly, in all research as we attempt to generate understanding about cause and effect relations from our structured observations.

Implication. When planning your studies and writing your papers should give clear explicit thought to the causal inferences; the methods explain the structure you have given to the observations you have made; the results what you observed; the discussion is where you explain why you believe what you have observed and that should be explicitly connected to the underlying causal assumptions. Yes, correlation does not imply causation, but causation implies correlation and causation is what is important to consider with your study—so tell the readers how confident you are and attempt to justify your confidence.

Two Premises

1. Explicitly outlined causal models provide us with a framework to iteratively build and share knowledge for practice and connects practice to research as well as research to practice.
2. Causal models provide frameworks upon which inference patterns can be taught, learned, practiced, and improved upon.

Three Examples

There are 3 features common to all 3 examples:

1. Causal models are being used as a representation of clinical knowledge and thus can be used to develop clinical reasoning
2. Causal models represent both causal structure and degrees of belief for research, practice, and education
3. Tools such as DAGitty (<http://www.dagitty.net/> [Retrieved July 21, 2018]) (freely available) and SAMIAM (<http://reasoning.cs.ucla.edu/samiam/> [Retrieved July 21, 2018]) (freely available) for the development, analysis, and use of physical therapy causal models in practice, research, and education

As presented in this article, based on the presentation at the CSM 2018, these examples are limited in depth and breadth, each of example could be an entire article.

SOCIAL DETERMINANTS OF HEALTH

Seeing a list of social factors that have been demonstrated to influence health (social determinants of health [SDH]) does not do justice to trying to understand the complexity of SDH. Figure 3A is a model that encodes the claims of a basic list. But this is not a great depiction of reality. Factors of SDH are causally connected to each other as well. One possible structure for those causal connections is in Figure 3B. The implications of these 2 structures on identifying through observational study or intervening through action are obvious. If Figure 3B represents reality, employment would have a leveraging impact given that it influences several other nodes in the causal structure. Such leverage effects would not be recognized if simply considering a list of SDH or even a model with the simplicity of Figure 3A.

DIFFERENTIAL DIAGNOSIS

Differential diagnosis imposes an abductive challenge. Abductive inference was first explained by Charles Pierce in the late 19th century.¹⁹ It is also described more completely as part of a critical realist philosophy of science by Bhaskar¹⁵ and referred to as retrodution. Abductive inference refers to any attempt to infer causes from observed effects (or more generally, antecedents from consequents). It is the general form of reasoning (inferential process) employed when attempting to diagnose, almost by definition since diagnosis is simply attempting to identify (through inference) the cause of observed effects (signs and symptoms). Even more so when signs are

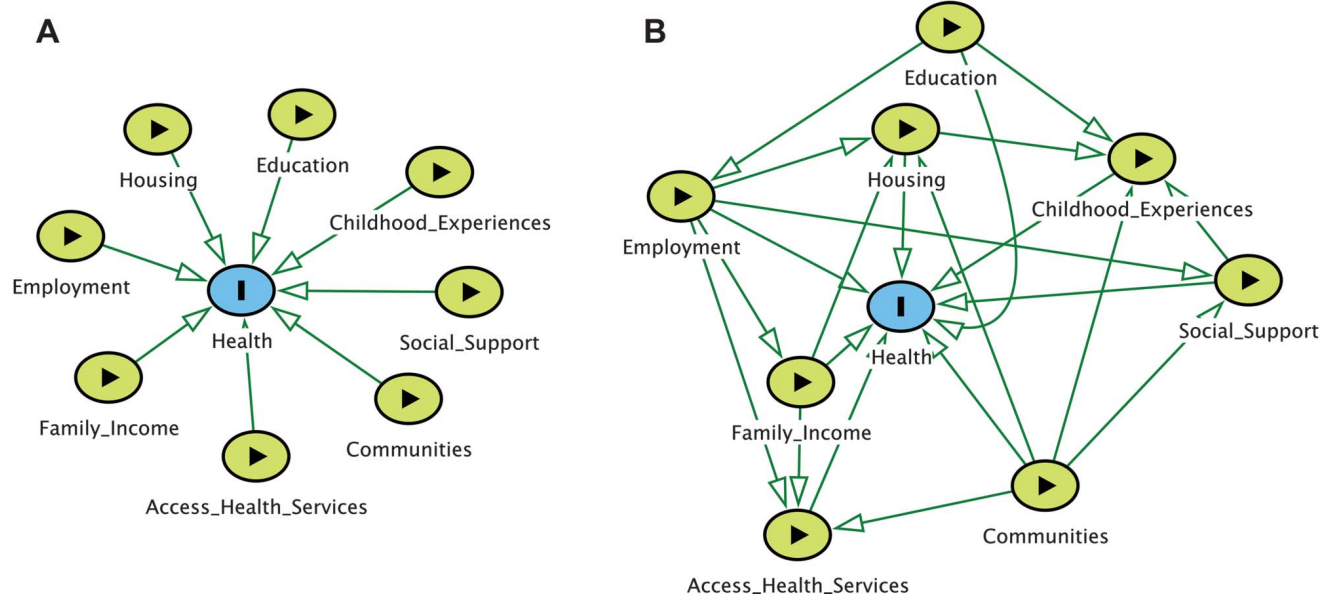


Fig. 3. Possible causal models of social determinants of health. A, Simple model—assumption that social determinants are independent of one another. B, Complex model—encodes possible relationships between social determinants.

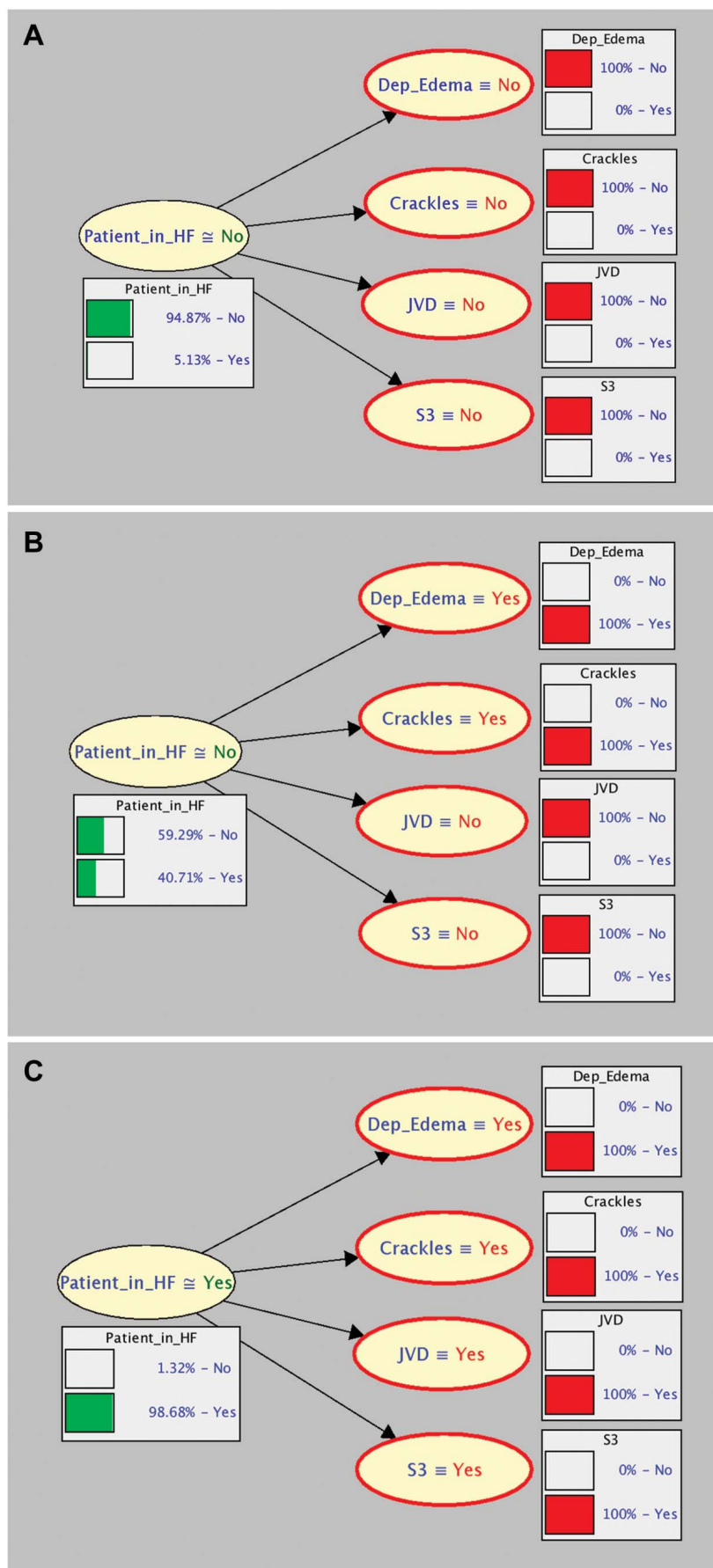


Fig. 4. Parameterized SAMIAM model of clinical signs of heart failure. S3, S3 heart sound; Dep Edema, dependent edema; JVD, jugular venous distension.

considered to include effects brought on by provocation (tests) or through technology with the goal of enhancing our sensory perceptual abilities (imaging, auscultation, electrodes). The abductive challenge is due to an inability to validate forms of inference as with deduction, and the fact of nonmutually exclusive causes.

There are at least 2 benefits to building causal models to assist with the differential diagnostic process. First, the ability to consider possible (alternative) causes, and second, the combination of diagnostic accuracy data (conditional probabilities) based on the mathematical structure of the causal graph. For this article, the focus will be on the second benefit. Figure 4 depicts 3 instantiations of a model built in SAMIAM (Model code and data available at: https://github.com/scollinspt/hf_cpg) based on diagnostic accuracy summary data for determining whether a patient presenting with dyspnea is in HF. The model is parameterized based on the reported sensitivity and specificity for each clinical sign (pulmonary crackles, dependent edema, jugular venous distention [JVD], and an S3 heart sound).^{20,21} Since reasoning through an actual case always includes being presented with more information than a single sign, the model allows for combining the information from the individual diagnostic accuracy data for testing various instantiations. In this case, if we consider each clinical sign as being present (Yes) or absent (No), there are $2^4 = 16$ possible combinations of the clinical signs (Table 1). Table 1 is arranged from the lowest probability of a patient being in HF to the highest probability. As anticipated, the 2 extremes are either all clinical signs absent ($P(\text{HF}) = 5.13\%$) or all clinical signs are present ($P(\text{HF}) = 98.68\%$). As anticipated, only presenting with one sign that has lower specificity such as dependent edema results in a low probability of being in HF ($P(\text{HF}) = 16.1\%$), whereas having one sign that has higher specificity such as an S3 heart sound results in a higher probability of being in HF ($P(\text{HF}) = 44.46\%$).

A benefit of such a model is that it is not static and can be updated with additional data. In fact, it can propose hypotheses of intercausal connections. For example, we expect that each of these clinical signs are correlated with one another due to their common physiological origin (congestion and elevated circulatory pressures due to a failing heart). Based on this model, the probability of an S3 heart sound increases from 3.2% when there is no dependent edema, JVD, or crackles, to 11.2% when there is dependent edema, JVD, and crackles. This is a model-based hypothesis that can be subjected to empirical study. The data from such a study can then be used to either validate or improve the model iteratively.

The benefit of iteratively building and using models as part of a practice, research, education cycle is the primary reason for my professional interest in the approach being advocated with causal models as part of KBP. If such an approach was advocated and adopted across the profession, then research would be designed and communicated with an underlying causal model in mind. Projects

could start with a model already in development or derive a model from an existing source or develop one based on reasons for proposing a new model. It would be a process for knowledge development within the profession that is based on a critical realist philosophy of science, and it would be at least one approach to answering the question I set out to answer in 2005 in my editorial on using complex systems methods to make clinical research more clinically relevant.¹⁴

We must ask ourselves, what came first, the nail or the hammer? Or perhaps the problem of fixing 2 pieces of wood was presented? The point is, with these tools available to us as a profession, how will we, or how should we make use of them for solving the problem? I do not believe that the gap between research and practice is going to get better by yelling “EVIDENCE-BASED PRACTICE” louder and louder. Decades have now passed. The problem remains. I do not believe that it is because clinicians are simply reluctant. I do not believe that it is simply that researchers don’t know how to generate useful protocols or results. But I do believe that you need the right tools to do a job. And you should take that from me, I do have a doctoral degree in ergonomics after all.

EFFECT SIZES TO CONDITIONAL PROBABILITIES

The last example to present is on the use of conditional probabilities to share findings with clinicians. There is little question that effect sizes are useful. They provide the clinician with an idea of the effect that is possible with an

TABLE 1
Estimated Probability of a Patient Being in Heart Failure Based on
a Model of Diagnostic Accuracy Data

Probability (Patient in HF) ^a	S3	Crackles	JVD	Edema
<i>0.051</i>	No	No	No	No
<i>0.161</i>	No	No	No	Yes
0.162	No	Yes	No	No
0.285	No	No	Yes	No
0.407	No	Yes	No	Yes
<i>0.445</i>	Yes	No	No	No
0.585	No	No	Yes	Yes
0.588	No	Yes	Yes	No
0.739	Yes	No	No	Yes
0.741	Yes	Yes	No	No
0.835	No	Yes	Yes	Yes
0.855	Yes	No	Yes	No
0.91	Yes	Yes	No	Yes
0.954	Yes	No	Yes	Yes
0.955	Yes	Yes	Yes	No
<i>0.987</i>	Yes	Yes	Yes	Yes

Italicized rows are specifically referred to in the text.
^aProbability reported from 0 to 1, with a probability of 1 = 100%.
HF, heart failure; JVD, jugular venous distention.

intervention (or from an exposure). Conditional probabilities would not obviate the need for effect sizes; in fact, effect sizes would be needed to estimate a conditional probability. There are at least 2 reasons for proposing the use of a conditional probability in addition to an effect size.

First, it is clinically useful. It is one thing to make the claim that in large samples of subjects, the effect size was a 63.5-m increase in 6MWD when using neuromuscular ES in patients with HF. However, it is also useful to say that the conditional probability of increasing 6MWD by at least the minimally important clinical difference of 30 m is 56%. A 56% probability of improvement is useful information and should not be compared to a null value of 50/50 because the chance of increasing 6MWD with no intervention would not be 50%. When considering ES as an intervention as part of a causal model, the connection $ES \rightarrow 6MWD$ is considered as the probability of increasing 6MWD ($6MWD_1$) given ES was completed (ES_1). In notational form, if ES effectively improves (causes) an increase in 6MWD, then

$$P(6MWD_1|ES_1) > P(6MWD_1|ES_0),$$

where ES_0 indicates that ES was not utilized (Data and estimates computed from the systematic review by Gomes Neto et al.¹⁷).

The second reasoning for proposing a conditional probability is that like differential diagnosis, it would provide the computational method to combine findings from several studies into a more complex causal model. The goal of which would be to address the problem described earlier in dealing with the empirical impossibility of conducting a study on 7 different interventions all with 4 possible approaches to intervention and the resultant $4^7 = 16,384$ required groups, and with 30 subjects per group, 491,520 subjects. That is just not empirically feasible. It is possible to get a better answer than we don't know that also does not require observing every possible scenario using well-constructed and iteratively updated and validated causal models.

The computational methods involved with computing a conditional probability and building a causal model that combines findings based on these probabilities are beyond the scope of this article (and were not reviewed in the presentation). They are a central part of manuscript in preparation.

CLOSING SUGGESTION

Let's use causal models to provide a synthesis of what we know to develop causal knowledge from which to

further develop with empirical evidence and from which knowledge we can practice. Causal models represent a synthesis of KBP with a critical realist epistemology that explicitly identifies the human mind as part of the process of knowledge and where the models that are constructed fit with reality even when we cannot empirically verify every component of such models; but where knowledge assumptions are encoded in the models clearly, combine knowledge with reasoning and are subject to empirical verification when possible.

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